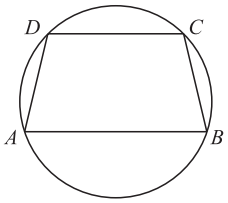
29. An isoceles trapezium is always cyclic.

Explanation A trapezium $ABCD$ in which $AB \parallel CD$ and $AD = BC$



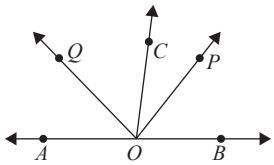
Then $ABCD$ is a cyclic trapezium.

EXERCISE-I

1. An angle is equal to one-third of its supplement. Its measure is equal to:
(a) 40° (b) 50°
(c) 45° (d) 55°

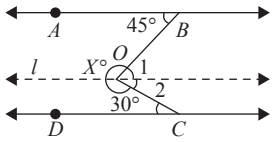
2. The complement of 30°20′ is:
(a) 69°40′ (b) 59°40′
(c) 35°80′ (d) 159°40′

3. In the given figure, OP bisect $\angle BOC$ and OQ bisects $\angle AOC$. Then $\angle POQ$ is equal to:



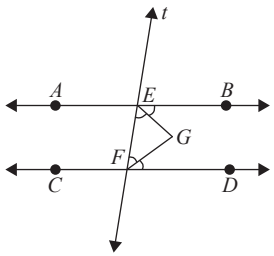
(a) 90° (b) 120°
(c) 60° (d) 100°

4. In the given, $AB \parallel CD$. Then X is equal to:



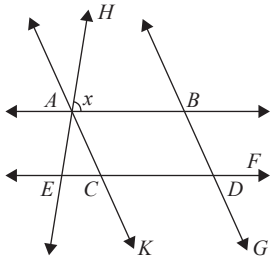
(a) 290° (b) 300°
(c) 280° (d) 285°

5. In the adjoining figure, $AB \parallel CD$, t is the transversal, EG and FG are the bisectors of $\angle BEE$ and $\angle DFE$ respectively, then $\angle EGF$ is equal to:



(a) 90° (b) 75°
(c) 80° (d) 110°

6. In the given figure, $AB \parallel CD$ and $AC \parallel BD$. If $\angle EAC = 40°$, $\angle FDG = 55°$, $\angle HAB = x$; then the value of x is:

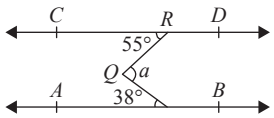


(a) 95° (b) 70°
(c) 35° (d) 85°

7. Find the measure of an angle, if six times its complement is 12° less than twice its supplement:
(a) 48° (b) 96°
(c) 24° (d) 58°

8. If two parallel lines are intersected by a transversal, then the bisectors of the two pairs of interior angles enclose a:
(a) Trapezium (b) Rectangle
(c) Square (d) none of these

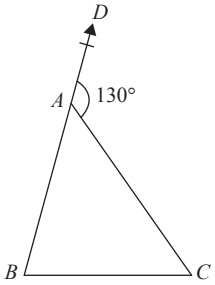
9. In fig., $AB \parallel CD$, $\angle a$ is equal to:



(a) 93° (b) 103°
(c) 83° (d) 97°

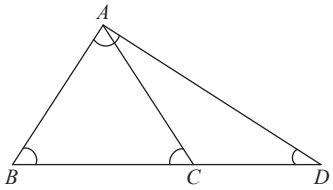
10. The complement of an angle exceeds the angle by 60°. Then the angle is equal to:
(a) 25° (b) 30°
(c) 15° (d) 35°

11. In the following figure, $\angle B:\angle C = 2:3$, find $\angle B + \angle C$.



(a) 120° (b) 52°
(c) 78° (d) 130°

12. In the given figure, $\angle B = \angle C = 55°$ and $\angle D = 25°$. Then:

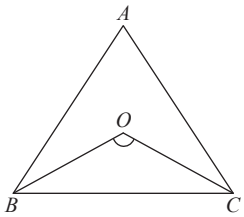


(a) $BC < CA < CD$
(b) $BC > CA > CA$
(c) $BC < CA$, $CA > CD$
(d) $BC > CA$, $CA < CD$

13. In a ΔABC, if $2\angle A = 3\angle B = 6\angle C$, Then $\angle A$ is equal to:
(a) 60° (b) 30°
(c) 90° (d) 120°

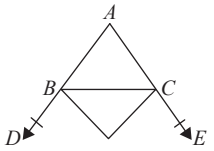
14. A, B, C are the three angles of a Δ. If $A - B = 15°$ and $B - C = 30°$. Then $\angle A$ is equal to:
(a) 65° (b) 80°
(c) 75° (d) 85°

15. In ΔABC, the angle bisectors of $\angle B$ and $\angle C$ meet at O. If $\angle A = 70°$, then $\angle BOC$ is equal to:



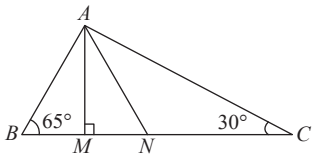
(a) 135° (b) 125°
(c) 115° (d) 110°

16. The sides AB and AC of ΔABC have been produced to D and E respectively. The bisectors of $\angle CBD$ and $\angle BCE$ meet at O. If $\angle A = 40°$, then $\angle BOC$ is equal to:

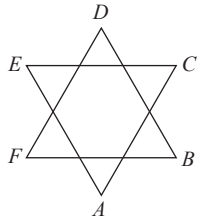


(a) 60° (b) 65°
(c) 75° (d) 70°

17. In the given figure, $AM \perp BC$ and AN is the bisector of $\angle A$. What is the measure of $\angle MAN$.
(a) 17.5° (b) 15.5°
(c) 20° (d) 25°

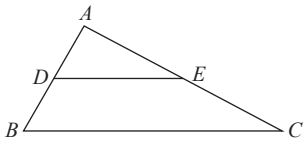


18. In the adjoining figure $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F =$



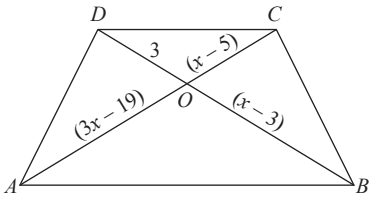
(a) 270° (b) 300°
(c) 360° (d) 330°

19. In the given figure, $DE \parallel BC$ if $AD = 1.7$ cm, $AB = 6.8$ cm and $AC = 9$ cm, find AE.



(a) 2.25 cm (b) 4.5 cm
(c) 1.25 cm (d) 2.5 cm

20. In the given figure, $AB \parallel DC$, find the value of x.

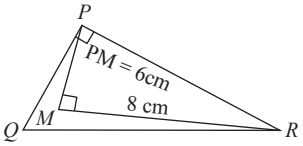


(a) $x = 8$ (b) $x = 9$
(c) $x = 8$ or 9 (d) $x = 10$

21. If the bisector of an angle of Δ bisects the opposite side, then the Δ is:

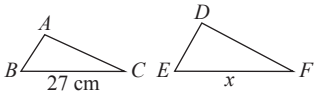
(a) Scalene (b) Isosceles
(c) Right triangle (d) None of these

22. In the given figure $\angle QPR = 90°$, $QR = 26$ cm, $PM = 6$ cm, $MR = 8$ cm and $\angle PMR = 90°$, find the area of ΔPQR.



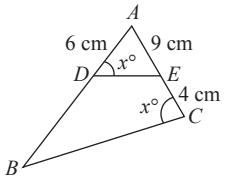
(a) 180 cm^2 (b) 240 cm^2
(c) 120 cm^2 (d) 150 cm^2

23. The areas of two similar Δs are 81 cm^2 and 144 cm^2. If the largest side of the smaller Δ is 27 cm, then the largest side of the larger Δ is:



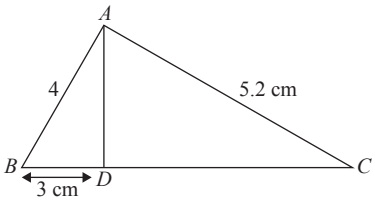
(a) 24 cm (b) 48 cm
(c) 36 cm (d) None of these

24. In the given figure, find the length of BD.



(a) 13.5 cm (b) 12 cm
(c) 14.5 cm (d) 15 cm

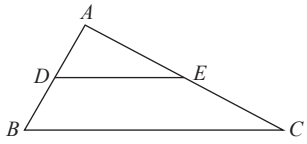
25. In the given figure $\angle BAD = \angle CAD$. $AB = 4$ cm, $AC = 5.2$ cm, $BD = 3$ cm. Find BC.



(a) 6.9 cm (b) 9.6 cm
(c) 3.9 cm (d) 9.3 cm

26. A ladder 15 m long reaches a window which is 9 m above the ground on one side of street. Keeping its foot at the same point, the ladder is turned to the other side of the street to reach a window 12 m high. What is the width of the street:
(a) 31 m (b) 12 m
(c) 30 m (d) 21 m

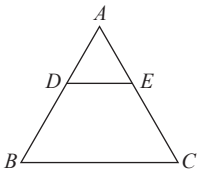
27. In, ΔABC, D and E are the mid-points of AB and AC respectively. Find the ratio of the areas of ΔADE and ΔABC.



(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $\frac{3}{4}$ (d) $\frac{1}{8}$

28. A vertical stick 12 cm long casts a shadow 8 cm long on the ground. At the same time a tower casts the shadow 40 m long on the ground. Find the height of the tower.
(a) 600 m (b) 160 m
(c) 60 m (d) 52 m

29. D and E are the points on the sides AB and AC respectively of ΔABC such that $AD = 8$ cm, $BD = 12$ cm, $AE = 6$ cm and $EC = 9$ cm. Then find BC/DE.



(a) $\frac{5}{2}$ (b) $\frac{2}{5}$
(c) $\frac{5}{7}$ (d) $\frac{5}{3}$

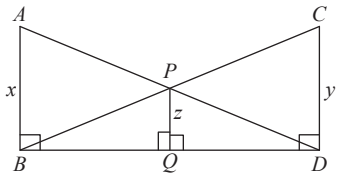
30. In an equilateral ΔABC, if $AD \perp BC$, then:
(a) $3AB^2 = 2AD^2$ (b) $2AB^2 = 3AD^2$
(c) $3AB^2 = 4AD^2$ (d) $4AB^2 = 3AD^2$

31. In a right angled ΔABC, rt. angled at A, $AD \perp BC$. Then:
(a) $AD^2 = BD \times CD$ (b) $AD^2 = AB \times AC$
(c) $AD^2 = BD \times AB$ (d) $AD^2 = CD \times AC$

32. If $ABCD$ is a ||gm and AC and BD be its diagonals, then:
(a) $AB^2 + BC^2 + CD^2 + DA^2 = AC^2 - BD^2$
(b) $AB^2 + BC^2 + CD^2 + DA^2 = AC^2 + BD^2$
(c) $4AD^2 = 2AC^2 + 2BD^2$
(d) $4AB^2 = 2AC^2 - 2BC^2$

33. In the given figure, $\angle ABD = \angle CBD = \angle PQB = 90°$. Then:

(a) $\frac{1}{x} - \frac{1}{y} = \frac{1}{z}$ (b) $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$
(c) $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$ (d) $\frac{1}{y} - \frac{1}{x} = \frac{1}{z}$

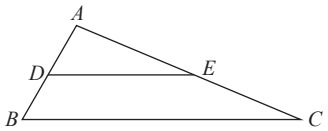


34. The area of two similar Δs are 121 cm^2 and 81 cm^2 respectively. What is the ratio of their corresponding heights (altitudes):

(a) $\frac{11}{9}$ (b) $\frac{22}{9}$

(c) $\frac{11}{18}$ (d) None of these

35. In the given figure, $DE \parallel BC$ and $DE : BC = 3:5$ the ratio of the areas of ΔADE and the trapezium $BCED$.



(a) $\frac{9}{25}$ (b) $\frac{12}{25}$

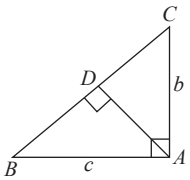
(c) $\frac{3}{4}$ (d) $\frac{9}{16}$

36. In a ΔABC, AD intersects ∠A and BC. If $BC = a$, $AC = b$ and $AB = c$, Then:

(a) $CD = \frac{b+c}{ab}$ (b) $CD = \frac{ab}{b+c}$

(c) $CD = \frac{bc+ab}{ac}$ (d) $CD = \frac{ac}{bc+ab}$

37. In the given figure, what is the length of AD in terms of b and c:



(a) $\frac{bc}{b^2+c^2}$ (b) $\frac{b^2+c^2}{bc}$

(c) $\frac{\sqrt{b^2+c^2}}{bc}$ (d) $\frac{bc}{\sqrt{b^2+c^2}}$

38. ABC is a Δ in which $AB = AC$ and D is a point on AC such that $BC^2 = AC \times CD$. Then:

(a) $BD = DC$ (b) $BD = BC$

(c) $BD = AB$ (d) $BD = AD$

39. Two poles of ht. a and b metres are p metres apart ($b > a$). The height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is:

(a) $\frac{a+b}{ab}$ (b) $\frac{p}{a+b}$

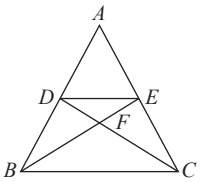
(c) $\frac{ab}{a+b}$ (d) $\frac{a+b}{p}$

40. ABC is a right Δ, right-angled at C. If $AB = c$, $BC = a$ and $CA = b$ and p is the length of the perpendicular from C on AB. Then:

(a) $\frac{1}{p^2}+\frac{1}{a^2}=\frac{1}{b^2}$ (b) $\frac{1}{p^2}+\frac{1}{b^2}=\frac{1}{a^2}$

(c) $\frac{1}{a^2}=\frac{1}{b^2}+\frac{1}{p^2}$ (d) $\frac{1}{p^2}=\frac{1}{a^2}+\frac{1}{b^2}$

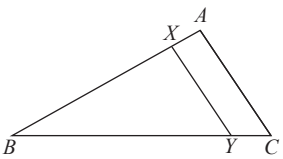
41. In the given figure $DE \| BC$ and $AD : DB = 5:4$, Then $\frac{ar(\Delta DFE)}{ar(\Delta CFB)}$



(a) $\frac{25}{81}$ (b) $\frac{25}{16}$

(c) $\frac{16}{25}$ (d) $\frac{16}{81}$

42. In the fig. $XY \| AC$ and XY divides triangular region ABC into two part equal in area. Then $\frac{AX}{AB}$ is equal to:



(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{\sqrt{2}+2}{\sqrt{2}}$

(c) $\frac{1}{2}$ (d) $\frac{\sqrt{2}-1}{\sqrt{2}}$

43. A point O in the interior of a rectangular $ABCD$ is joined with each of the vertices A, B, C and D. Then:

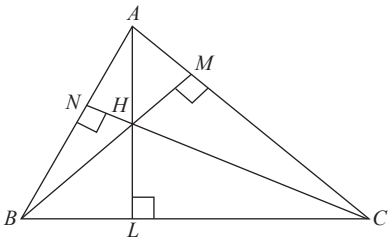
(a) $OA^2 + OC^2 = OB^2 - OD^2$

(b) $OA^2 + OC^2 = OB^2 + OD^2$

(c) $OA^2 = OB^2 = OC^2 + OD^2$
(d) $OA^2 + OD^2 = OB^2 + OC^2$

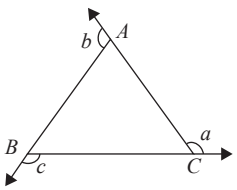
44. In ΔABC, the median BE intersects AC at E, if BG = 6 cm, where G is the centroid, then BE is equal to:
(a) 7 cm (b) 9 cm
(c) 8 cm (d) 10 cm

45. If H is the orthocentre of ΔABC, then the orthocentre of ΔHBC is (fig. given):



(a) N (b) M
(c) A (d) L

46. If the sides of a triangle are produced then the sum of the exterior angles i.e., $\angle a + \angle b + \angle c$ is equal to:



(a) 180° (b) 90°
(c) 360° (d) 270°

47. Incentre of a triangle lies in the interior of:
(a) an isosceles triangle only
(b) any triangle
(c) an equilateral triangle only
(d) a right triangle only

48. In a ΔABC, the bisectors of $\angle B$ and $\angle C$ *intersect* each other at apoint O. Then $\angle BOC$ is equal to:

(a) $90° - \frac{1}{2}\angle A$ (b) $120° + \frac{1}{2}\angle A$

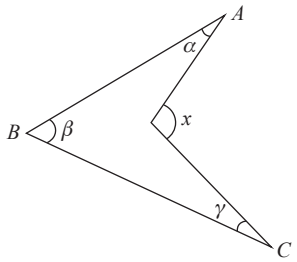
(c) $90° + \frac{1}{2}\angle A$ (d) $120° - \frac{1}{2}\angle A$

49. In a ΔABC, the sides AB and AC are produced to P and Q respectively. The bisectors of $\angle OBC$ and $\angle QCB$ intersect at a point O. Then $\angle BOC$ is equal to:

(a) $90° + \frac{1}{2}\angle A$ (b) $90° - \frac{1}{2}\angle A$

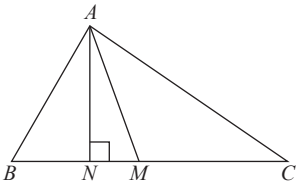
(c) $120° + \frac{1}{2}\angle A$ (d) $120° - \frac{1}{2}\angle A$

50. In the given figure, which of the following is true:



(a) $x = \alpha + \beta + \gamma$ (b) $x + \beta = \alpha + \gamma$
(c) $x + \gamma = \beta + \alpha$ (d) $x + \alpha = \beta + \gamma$

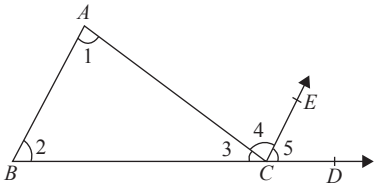
51. In the given figure, In a ΔABC, $\angle B = \angle C$. If AM is the bisector of $\angle BAC$ and $AN \perp BC$, then $\angle MAN$ is equal to:



(a) $\frac{1}{2}(\angle B + \angle C)$ (b) $\frac{1}{2}(\angle C - \angle B)$

(c) $\angle B + \angle C$ (d) $\frac{1}{2}(\angle B - \angle C)$

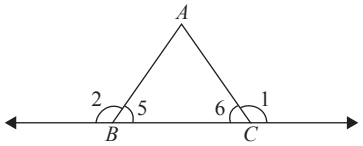
52. In the given figure, side BC of ΔABC is produced to form ray BD and $CE \parallel BA$. Then $\angle ACD$ is equal to:



(a) $\angle A - \angle B$ (b) $\frac{1}{2}(\angle A + \angle B)$

(c) $\angle A + \angle B$ (d) $\frac{1}{2}(\angle A - \angle B)$

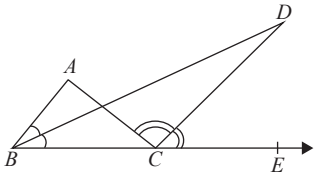
53. In the given figure, the side BC of a ΔABC is produced on both sides. Then $\angle 1 + \angle 2$ is equal to:



(a) $\angle A + 180°$ (b) $180° - \angle A$

(c) $\frac{1}{2}(\angle A + 180°)$ (d) $\angle A + 90°$

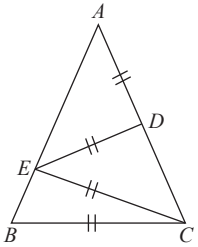
54. In the figure, BD and CD are angle bisectors of $\angle ABC$ and $\angle ACE$, respectively. Then $\angle BDC$ is equal to:



(a) $\angle BAC$ (b) $2\angle BAC$

(c) $\frac{1}{2}\angle BAC$ (d) $\frac{1}{3}\angle BAC$

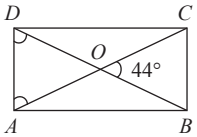
55. In fig, $AB = AC$, D is a point on AC and E on AB such that $AD = ED = EC = BC$. Then $\angle A:\angle B$:



(a) 1:2 (b) 2:1

(c) 3:1 (d) 1:3

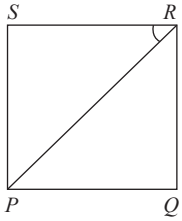
56. The diagonals of a rectangle $ABCD$ meet at 0. If $\angle BOC = 44°$, then $\angle OAD$ is equal to:



(a) 90° (b) 60°

(c) 100° (d) 68°

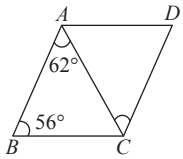
57. $PQRS$ is a square. The $\angle SRP$ is equal to:



(a) 45° (b) 90°

(c) 100° (d) 60°

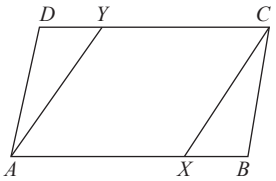
58. $ABCD$ is a rhombus with $\angle ABC = 56°$, then $\angle ACD$ is equal to:



(a) 90° (b) 60°

(c) 56° (d) 62°

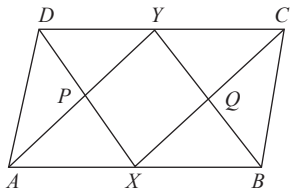
59. $ABCD$ is a parallelogram and X, Y are the mid-points of sides AB and CD respectively. Then quadrilateral $AXCY$ is a:



(a) parallelogram (b) rhombus

(c) square (d) rectangle

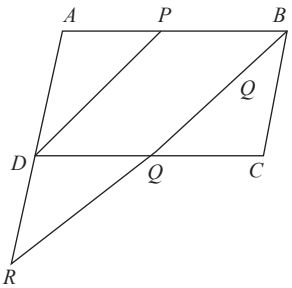
60. X, Y are the mid-points of opposite sides AB and DC of a parallelogram $ABCD$. AY and DX are joined intersecting in P; CX and BY are joined intersecting in Q. Then $PXQY$ is a:



(a) Rectangle (b) Rhombus

(c) Parallelogram (d) Square

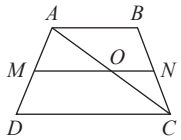
61. P is the mid-point of side AB to a parallelogram $ABCD$. A line through B parallel to PD meets DC at Q and AD produced at R. Then BR is equal to:



(a) BQ (b) $\frac{1}{2}$

(c) $2BQ$ (d) None of these

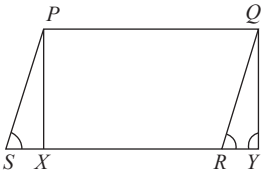
62. $ABCD$ is a trapezium in which $AB \parallel CD$. M and N are the mid-points of AD and respectively. If AB = 12 cm an MN = 14 cm. Find CD.



(a) 2 cm (b) 5 cm

(c) 12 cm (d) 16 cm

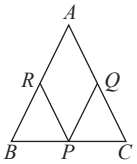
63. $PQRS$ is a parallelogram. PX and QY are, respectively, the perpendicular form P and Q to SR and SR produced. The PX is equal to:



(a) QY (b) $2QY$

(c) $\frac{1}{2}Q$ (d) XR

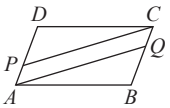
64. In a ΔABC, P, Q and R are the mid-points of sides BC, CA and AB respectively. If AC = 21 cm, BC = 29 cm and AB = 30 cm. The perimeter of the quad. $ARPQ$ is:



(a) 91 cm (b) 60 cm

(c) 51 cm (d) 70 cm

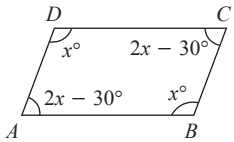
65. $ABCD$ is a parallelogram. P is a point on AD such that $AP = \frac{1}{3}AD$ and Q is a point on BC such that $CQ = \frac{1}{3}BC$. Then $AQCP$ is a:



(a) Parallelogram (b) Rhombus

(c) Rectangle (d) Square

66. Find the measure of each angle of a parallelogram, if one of its angles is 30° less than twice the smallest angle.



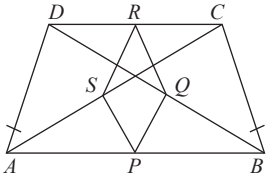
(a) 60°, 100°, 90°, 20°

(b) 80°, 40°, 120°, 90°

(c) 100°, 90°, 90°, 80°

(d) 70°, 110°, 70°, 110°

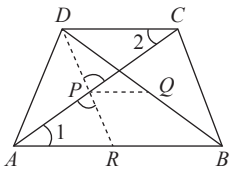
67. $ABCD$ is a trapezium in which $AB \parallel DC$ and $AD = BC$. If P, Q, R, S be respectively the mid-point of BA, BD and CD, CA. The $PQRS$ is a:



(a) Rhombus (b) Rectangle

(c) Parallelogram (d) Square

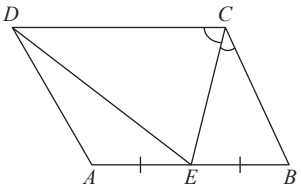
68. $ABCD$ is a trapezium and P, Q are the mid-points of the diagonals AC and BD. Then PQ is equal to:



(a) $\frac{1}{2}(AB)$ (b) $\frac{1}{2}(CD)$

(c) $\frac{1}{2}(AB - CD)$ (d) $\frac{1}{2}(AB + CD)$

69. $ABCD$ is a parallelogram, E is the mid-point of AB and CE bisects $\angle BCD$. The $\angle DEC$ is:



(a) 60° (b) 90°

(c) 100° (d) 120°

70. The angles of a quadrilateral are respectively 100°, 98° and 92°. The fourth angle is equal to:

(a) 90° (b) 95°

(c) 70° (d) 75°

71. The measure of each angle of a regular hexagon is:

(a) 110° (b) 130°

(c) 115° (d) 120°

72. The interior angle of a regular polygon is 108°. The number of sides of the polygon is:

(a) 6 (b) 7

(c) 8 (d) 5

73. The number of diagonals in a hexagon is:

(a) 9 (b) 8

(c) 10 (d) 7

74. If the number of diagonals of a polygon is 27, then the number of sides is:

(a) 10 (b) 9

(c) 11 (d) 6

75. One angle of a pentagon is 140°. If the remaining angles are in the ratio 1:2:3:4. The size of the greatest angle is:

(a) 150° (b) 180°

(c) 160° (d) 170°

76. The exterior angle of a regular polygon is $\frac{1}{3}$ of its interior angle. The number of the sides of the polygon is:
(a) 9 (b) 8
(c) 10 (d) 12

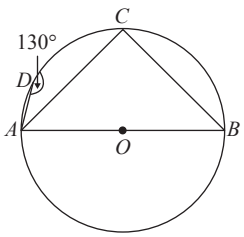
77. The ratio of the measure of an angle of a regular octagon to the measure of its exterior angle is:
(a) 3:1 (b) 2:1
(c) 1:3 (d) 1:2

78. $ABCDE$ is a regular pentagon. Diagonal AD divides $\angle CDE$ in to parts, then the ratio of $\frac{\angle ADE}{\angle ADC}$ is equal to:
(a) 3:1 (b) 1:4
(c) 1:3 (d) 1:2

79. The difference between an exterior angle of $(n-1)$ sided regular polygon and an exterior angle of $(n-2)$ sided regular polygon is 6°, then the value of n is:
(a) 14 (b) 15
(c) 13 (d) 12

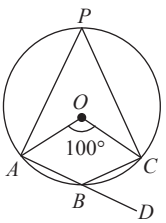
80. The radius of a circle is 13 cm and the length of one of its chords is 10 cm. What is the distance of the chord from the centre:
(a) 10 cm (b) 15 cm
(c) 12 cm (d) 9 cm

81. In the given figure, $ABCD$ is a cyclic quadrilateral whose side AB is a diameter of the circle through A, B and C. If $\angle ADC = 130°$, find $\angle CAB$.



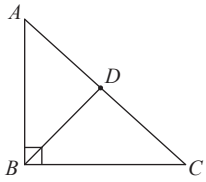
(a) 40° (b) 50°
(c) 30° (d) 130°

82. In the given figure, O is the centre of the circle, Find $\angle CBD$.



(a) 140° (b) 50°
(c) 40° (d) 130°

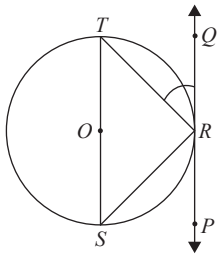
83. In a ΔABC, $\angle$B is a right angle, $AC = 6$ cm and D is the mid-point of AC. Find the length of BD.



(a) 4 cm (b) $\sqrt{6}$ cm
(c) 3 cm (d) 4.5 cm

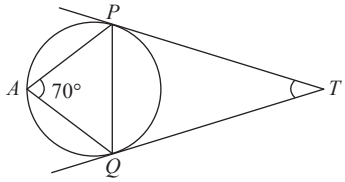
84. A cyclic quadirateral whose opposite angles are equal, is a:
(a) Parallelogram but not a rhombus
(b) Rhombus
(c) Rectangle
(d) Square but not a rectangle

85. In the given figure, ST is a diameter of the circle with centre O and PQ is the tangent at a point R. If $\angle TRQ = 40$, what is $\angle RTS$:



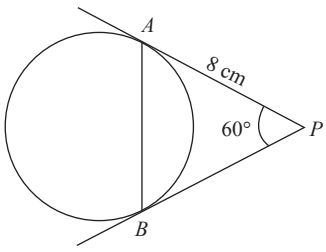
(a) 40° (b) 50°
(c) 60° (d) 30°

86. In the given figure, TP and TQ are tangents to the circle. If $\angle PAQ = 70°$, what is $\angle PTQ$?



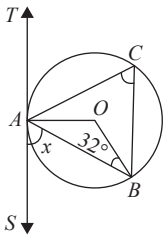
(a) 30° (b) 45°
(c) 60° (d) 40°

87. In the given figure, PA and PB are tangents from a point P to a circle such that $PA = 8$ cm and $\angle APB = 60°$. What is the length of the chord AB?



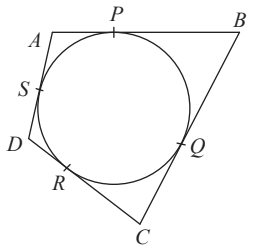
(a) 8 cm (b) 10 cm
(c) 6 cm (d) 12 cm

88. In the given figure, *TAS* is a tangent to the circle at the point *A*. If $\angle OBA = 32°$, what is the value of *x*:



(a) 64° (b) 40°
(c) 58° (d) 50°

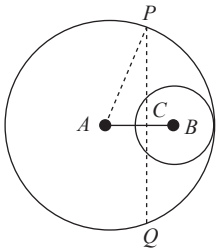
89. In the given figure, a circle touches all the four sides of quadrilateral *ABCD* whose sides *AB* = 6 cm, *BC* = 7 cm and *CD* = 4 cm. Find *AD*.



(a) 5 cm (b) 4 cm
(c) 3 cm (d) 2 cm

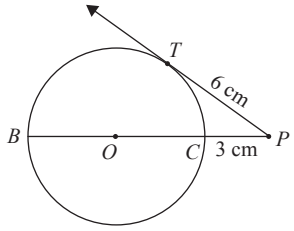
90. *AB* and *CD* are two parallel chords of a circle such that *AB* = 10 cm and *CD* = 24 cm. If the chords are on opposite sides of the centre and the distance between them is 17 cm, what is the radius of the circle:
(a) 14 cm (b) 10 cm
(c) 13 cm (d) 15 cm

91. In the given figure, two circle with centres *A* and *B* of radii 5 cm and 3 cm touch each other internally. If the perpendicular bisector of segment *AB* meets the bigger circle in *P* and *Q*, find the length of *PQ*.



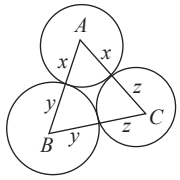
(a) $4\sqrt{6}$ cm (b) $\sqrt{24}$ cm
(c) $8\sqrt{3}$ cm (d) $4\sqrt{3}$ cm

92. In the given figure *O* is the centre of the circle and *PT* is a tangent at *T*. If *PC* = 3cm, *PT* = 6 cm, calculate the radius of the circle.



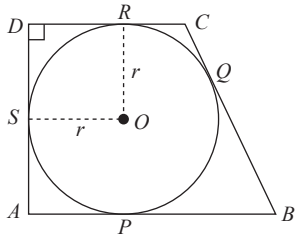
(a) 9 cm (b) 4.5 cm
(c) 8 cm (d) 12 cm

93. With the vertices of a ΔABC as centres, three circles are described, each touching the other two externally. If the sides of the Δ are 9 cm, 7 cm and 6 cm, find the radii of the circles:



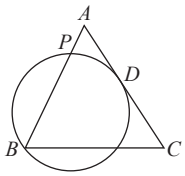
(a) 4cm, 7cm and 3cm
(b) 7cm, 5cm and 2cm
(c) 5cm, 4cm and 3cm
(d) 4cm, 5cm and 2cm

94. In the given figure, *ABCD* is a quadrilateral in which $\angle O = 90$. A circle *C*(0, *r*) touches the sides *AB*, *BC*, *CD* and *DA* at *P*, *Q*, *R*, *S* respectively. If *BC* = 38 cm, *CD* = 25 cm and *BP* = 27 cm, find the value of *r*.



(a) 14 cm (b) 15 cm
(c) 10 cm (d) 16 cm

95. In the given figure, *ABC* is an isosceles Δ in which *AB* = *AC*. A circle through *B* touches *AC* at its mid-point *D* and intersects *AB* at *P*. Then which of the following is correct:



(a) $AP = \frac{3}{4}AB$ (b) $AP = \frac{2.5}{3}AB$
(c) $AP = \frac{4}{5}AB$ (d) $AP = \frac{1}{4}AB$

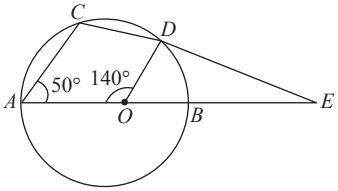
96. ABC is a right angled Δ with $BC = 6$ cm and $AB = 8$ cm. A circle with centre O is inscribed in ΔABC. The radius of the circle is:

(a) 4 cm (b) 3 cm
(c) 2 cm (d) 1 cm

97. If all the sides of a parallelogram touch a circle, then the parallelogram is a:

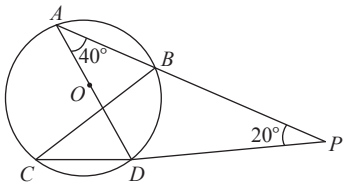
(a) Rectangle (b) Rhombus
(c) Square (d) None

98. In the given figure, O is the centre of a circle. If $\angle AOD = 140°$ and $\angle CAB = 50°$, what is $\angle EDB$:



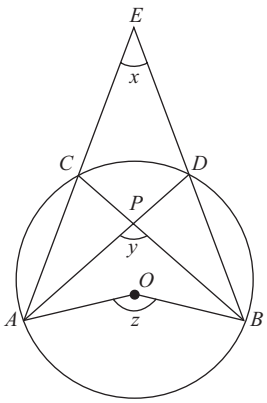
(a) 70° (b) 40°
(c) 60° (d) 50°

99. PBA and PDC are two secants. AD is the diameter of the circle with centre at O. $\angle A = 40$, $\angle P = 20°$. Find the measure of $\angle DBC$.



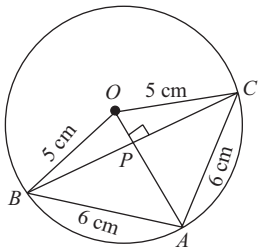
(a) 30° (b) 45°
(c) 50° (d) 40°

100. In the given figure, O is the centre of the circle. Then $\angle x + \angle y$ is equal to:



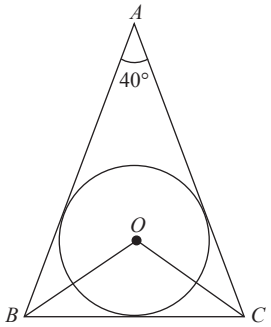
(a) $2\angle z$ (b) $\frac{\angle z}{2}$
(c) $\angle z$ (d) None

101. In a circle of radius 5 cm, AB and AC are two equal chords such that $AB = AC = 6$ cm. What is the length of the chord BC.



(a) 9.6 cm (b) 11 cm
(c) 12 cm (d) 9 cm

102. In the given figure O is the centre of incircle for ΔABC. Find $\angle BOC$ if $\angle BAC = 40°$.



(a) 100° (b) 120°
(c) 90° (d) 110°

103. If an equilateral triangle ABC be inscribed in a circle, then the tangents at their vertices will form another Δ.

(a) scalene (b) equilateral
(c) Isosceles (d) Right Δ

104. The radius of the incircle of a Δ is 4 cm and the segments into which one side is divided by the point of contact are 6 cm and 8 cm, then the length of the shortest side of the Δ is:

(a) 12 cm (b) 15 cm
(c) 13 cm (d) 14 cm

EXERCISE-2
(BASED ON MEMORY)

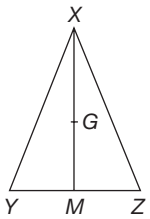
1. If the height of the equilateral triangle is $2\sqrt{3}$ cm, then determine the area (in cm^2) of the equilateral triangle.

(a) 6 (b) $2\sqrt{3}$

(c) $4\sqrt{3}$ (d) 12

[SSC CHSL (10+2) Tier-I CBE, 2018]

2. In triangle XYZ, G is the centroid. If XY = 11cm, YZ = 14cm and XZ = 7 cm, then what is the value (in cm) of GM?

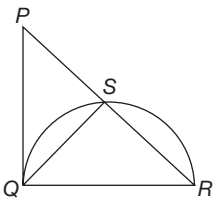


(a) 6 (b) 4

(c) 2 (d) 3

[SSC CGL Tier-II CBE, 2018]

3. In the given figure, triangle PQR is a right angled triangle at Q. If PQ = 35 cm and QS = 28 cm, then what is the value (in cm) of SR?

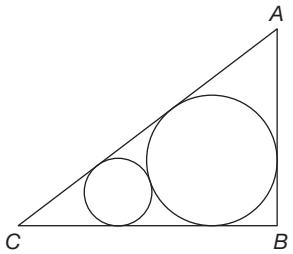


(a) 35.33 (b) 37.33

(c) 41.33 (d) 43.33

[SSC CGL Tier-I CBE, 2018]

4. In the given figure, ABC is a right angled triangle. $\angle ABC = 90°$ and $\angle ACB = 60°$. If the radius of the smaller circle is 2 cm, then what is the radius (in cm) of the larger circle?



(a) 4 (b) 6

(c) 4.5 (d) 7.5

[SSC CGL Tier-II CBE, 2018]

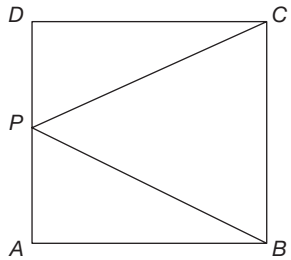
5. PQRS is a square whose side is 16 cm. What is the value of the side (in cm) of the largest regular octagon that can be cut from the given square?

(a) $8-4\sqrt{2}$ (b) $16+8\sqrt{2}$

(c) $16\sqrt{2}-\sqrt{16}$ (d) $16-\sqrt{2}$

[SSC CGL Tier-II CBE, 2018])

6. In the given figure, ABCD is a square whose side is 4 cm. P is a point on the side AD. What is the minimum value (in cm) of BP + CP?



(a) $4\sqrt{5}$ (b) $4\sqrt{4}$

(c) $6\sqrt{3}$ (d) $4\sqrt{6}$

[SSC CGL Tier-II CBE, 2018])

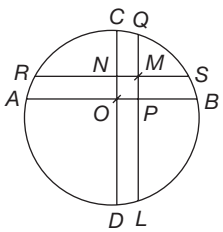
7. If a regular polygon has 5 sides then the measure of its interior angle is greater than the measure of its exterior angle by how many degrees?

(a) 60° (b) 36°

(c) 90° (d) 100°

[SSC CHSL (10+2) Tier-I CBE, 2018]

8. In the given figure, CD and AB are diameters of circle and AB and CD are perpendicular to each other. LQ and SR are perpendiculars to AB and CD respectively. Radius of circle is 5 cm, PB:PA = 2:3 and CN:ND = 2:3. What is the length (in cm) of SM?

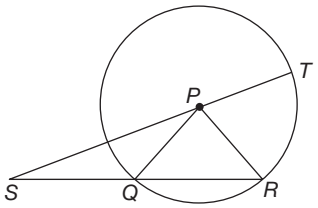


(a) $\left[(5\sqrt{3})-3\right]$ (b) $\left[(4\sqrt{3})-2\right]$

(c) $\left[(2\sqrt{5})-1\right]$ (d) $\left[(2\sqrt{6})-1\right]$

[SSC CGL Tier-II CBE, 2018]

9. In the given figure, P is the centre of the circle. If QS = PR, then what is the ratio of ∠RSP to the ∠TPR?

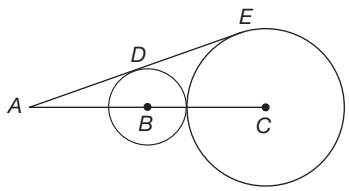


(a) 1:4 (b) 2:5

(c) 1:3 (d) 2:7

[SSC CGL Tier-II CBE, 2018]

10. In the given figure, B and C are the centres of the two circles. ADE is the common tangent to the two circles. If the ratio of the radius of both the circles is 3:5 and AC = 40, then what is the value of DE?



(a) $3\sqrt{15}$ (b) $5\sqrt{15}$

(c) $6\sqrt{15}$ (d) $4\sqrt{15}$

[SSC CGL Tier-II CBE, 2018])

11. The radius of two circles is 3 cm and 4 cm. The distance between the centres of the circles is 10 cm. What is the ratio of the length of direct common tangent to the length of the transverse common tangent?

(a) $\sqrt{51}:\sqrt{68}$ (b) $\sqrt{33}:\sqrt{17}$

(c) $\sqrt{66}:\sqrt{51}$ (d) $\sqrt{28}:\sqrt{17}$

[SSC CGL Tier-II CBE, 2018])

12. ΔABC is equilateral triangle. O is the point of intersection of altitudes AL, BM and CN. If OA = 16 cm, what is the semi-perimeter (in cm) of the triangle ABC?

(a) $8\sqrt{3}$ (b) $12\sqrt{3}$

(c) $16\sqrt{3}$ (d) $24\sqrt{3}$

[SSC CAPFs ASI & Delhi Police SI, 2017]

13. DEF is an isosceles triangle such that DE = DF = 60 cm and EF = 96 cm. DG is a median to base EF. What is the length (in cm) of DG?

(a) 22 (b) 36

(c) 24 (d) 32

[SSC CAPFs ASI & Delhi Police SI, 2017]

14. The side BC of ΔABC is produced to D. If ∠ACD = 114° and $\angle ABC = \left(\frac{1}{2}\right)\angle BAC$, what is the value (in degrees) of ∠BAC?

(a) 36 (b) 48

(c) 76 (d) 84

[SSC CGL Tier-I (CBE), 2017]

15. In ΔPQR, ∠P:∠Q:∠R = 1:3:5. What is the value (in degrees) of ∠R − ∠P?

(a) 30 (b) 80

(c) 45 (d) 60

[SSC CAPFs ASI & Delhi Police SI, 2017]

16. In triangle PQR, the sides PQ and PR are produced to A and B respectively. The bisectors of ∠AQR and ∠BRQ intersect at point O. If ∠QOR = 50, what is the value (in degree) of ∠QPR?

(a) 50 (b) 60

(c) 80 (d) 100

[SSC CGL Tier-I (CBE), 2017]

17. In a triangle PQR, ∠Q = 90°. If PQ = 12 cm QR = 5 cm, then what is the radius (in cm) of the circumcircle of the triangle?

(a) 5 (b) 6

(c) 6.5 (d) $6\sqrt{2}$

[SSC CGL Tier-I CBE, 2017]

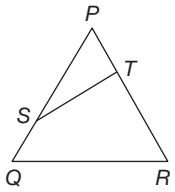
18. PQRA is a rectangle, AP = 22 cm, PQ = 8 cm. ΔABC is a triangle whose vertices lie on the sides of PQRA such that BQ = 2 cm and QC = 16 cm. Then the length of the line joining the mid points of the sides AB and BC is

(a) $4\sqrt{2}$ cm (b) 5 cm

(c) 6 cm (d) 10 cm

[SSC CGL Tier-II (CBE), 2017]

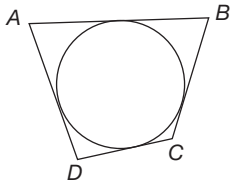
19. In the given figure, QRTS is a cyclic quadrilateral. If PT = 5 cm, SQ = 4 cm, PS = 6 cm and ∠PQR = 63°, then what is the value (in cm) of TR?



(a) 3 (b) 7
(c) 9 (d) 15

[SSC CHSL Tier-I CBE, 2017]

20. In the given figure, a circle touches quadrilateral ABCD. If AB = $2x + 3$, BC = $3x - 1$. CD = $x + 6$ and DA = $x + 4$, then what is the value of x?



(a) 3 (b) 4.5
(c) 6 (d) 6.5

[SSC CGL Tier-I (CBE), 2017]

21. The lengths of the sides of a triangle area a, b and c respectively. If $a^2 + b^2 + c^2 = ab + bc + ca$, then the triangle is:

(a) isosceles (b) equilateral
(c) scalene (d) right-angled

[SSC CGL Tier-I (CBE), 2016]

22. BD and CE are two medians of the triangle ABC. If EO = 7 cm. then the length of CE is

(a) 28 cm (b) 14 cm
(c) 21 cm (d) 35 cm

[SSC CGL Tier-I (CBE), 2016]

23. The lengths of side AB and side BC of a scalene triangle ABC are 12 cm and 8 cm respectively. The size of angle C is 90°. Find the approximate length of side AC.

(a) 12 (b) 9
(c) 14 (d) 16

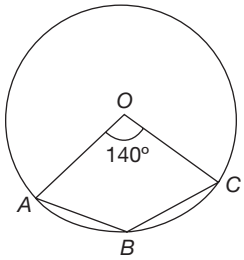
[SSC CGL Tier-I (CBE), 2016]

24. ΔABC is a triangle. PQ is line segment intersecting AB in P and AC in Q and PQ || BC. The ratio of AP:BP = 3:5 and length of PQ is 18 cm. The length of BC is

(a) 28 cm (b) 48 cm
(c) 84 cm (d) 42 cm

[SSC CGL Tier-II (CBE), 2016]

25. In the adjoining figure ∠AOC = 140 where O is the centre of the circle then ∠ABC is equal to:



(a) 110° (b) 100°
(c) 90° (d) 40°

[SSC CAPFs (CPO) SI & ASI, Delhi Police, 2016]

26. A square is inscribed in a quarter-circle in such a manner that two of its adjacent vertices lie on the two radii at an equal distance from the centre, while the other two vertices lie on the circular arc. If the square has sides of length x, then the radius of the circle is

(a) $\frac{16x}{\text{ð} + 4}$ (b) $\frac{2x}{\sqrt{\text{ð}}}$
(c) $\frac{\sqrt{5x}}{\sqrt{2}}$ (d) $\sqrt{2}x$

[SSC, 2015]

27. The measure of an angle whose supplement is three times as large as its complement is

(a) 30° (b) 45°
(c) 60° (d) 75°

[SSC, 2015]

28. Two chords of length a unit and b unit of a circle make angles 60° and 90° at the centre of a circle respectively, then the correct relation is

(a) $b = \sqrt{2}a$ (b) b = 2a
(c) $b = \sqrt{3}a$ (d) $b = \frac{3}{2}a$

[SSC, 2015]

29. In a parallelogram PQRS, angle P is four times of angle Q, then the measure of ∠R is

(a) 36° (b) 72°
(c) 130° (d) 144°

[SSC, 2015]

30. A tangent is drawn to a circle of radius 6 cm from a point situated at a distance of 10 cm from the centre of the circle. The length of the tangent will be

(a) 4 cm (b) 5 cm
(c) 8 cm (d) 7 cm

[SSC, 2015]

31. A ship after sailing 12 km towards south from a particular place covered 5 km more towards east. Then the straightway distance of the ship from that place is
(a) 18 km (b) 15 km
(c) 13 km (d) 11 km

[SSC, 2015]

32. A and B are centres of two circles of radii 11 cm and 6 cm, respectively. PQ is a direct common tangent to the circles. If $\overline{AB}$ = 13 cm, then length of $\overline{PQ}$ will be
(a) 8.5 cm (b) 17 cm
(c) 12 cm (d) 13 cm

[SSC, 2015]

33. Given that the ratio of altitudes of two triangles is 4 : 5, ratio of their areas is 3 : 2. The ratio of their corresponding bases is
(a) 5 : 8 (b) 15 : 8
(c) 8 : 5 (d) 8 : 15

[SSC, 2015]

34. AB and CD are two parallel chords of a circle of lengths 10 cm and 4 cm respectively. If the chords are on the same side of the centre and the distance between them is 3 cm, then the diameter of the circle is
(a) $2\sqrt{21}$ cm (b) $2\sqrt{29}$ cm
(c) $\sqrt{21}$ cm (d) $\sqrt{29}$ cm

[SSC, 2015]

35. In triangle ABC, DE || BC where D is a point on AB and E is a point on AC. DE divides the area of Δ ABC into two equal parts. Then DB : AB is equal to
(a) $(\sqrt{2}-1):\sqrt{2}$ (b) $(\sqrt{2}+1):\sqrt{2}$
(c) $\sqrt{2}:(\sqrt{2}+1)$ (d) $\sqrt{2}:(\sqrt{2}-1)$

[SSC, 2015]

36. If O is the circumcentre of a triangle ABC lying inside the triangle, the ∠OBC + ∠BAC is equal to
(a) 110º (b) 90º
(c) 60º (d) 120º

[SSC, 2015]

37. The interior angle of a regular polygon exceeds its exterior angle by 108º. The number of sides of the polygon is
(a) 14 (b) 12
(c) 16 (d) 10

[SSC, 2015]

38. In ΔABC, ∠BAC = 90° and AD ⊥ BC. If BD = 3 cm and CD = 4 cm, then the length (in cm) of AD is
(a) 6 (b) $2\sqrt{3}$
(c) 3.5 (d) 5

[SSC, 2015]

39. AD is perpendicular to the internal bisector of ∠ABC of ΔABC. DE is drawn through D and parallel to BC to meet AC at E. If the length of AC is 12 cm, then the length of AE (in cm.) is
(a) 6 (b) 8
(c) 3 (d) 4

[SSC, 2015]

40. The centroid of a Δ ABC is G. The area of Δ ABC is 60 cm^2. The area of ΔGBC is
(a) 10 cm^2 (b) 20 cm^2
(c) 40 cm^2 (d) 30 cm^2

[SSC, 2015]

41. Quadrilateral ABCD is circumscribed about a circle. If the lengths of AB, BC, CD are 7 cm, 8.5 cm and 9.2 cm respectively, then the length (in cm) of DA is
(a) 10.7 (b) 16.2
(c) 7.2 (d) 7.7

[SSC, 2015]

42. The measures of two angles of a triangle is in the ratio 4:5. If the sum of these two measures is equal to the measure of the third angle. Find the smallest angle.
(a) 50 (b) 10
(c) 40 (d) 90

[SSC, 2015]

43. If PQRS is a rhombus and ∠SPQ = 50°, then ∠RSQ is:
(a) 45° (b) 55°
(c) 65° (d) 75°

[SSC, 2015]

44. If two supplementary angles differ by 44°, then one of the angle is:
(a) 65° (b) 68°
(c) 72° (d) 102°

[SSC, 2015]

45. If two medians BE and CF of a triangle ABC, intersect each other at G and if BG = CG, ∠BGC = 60°, BC = 8cm, then area of the triangle ABC is:

(a) $48\sqrt{3}$ cm^2 (b) 48 cm^2

(c) $64\sqrt{3}$ cm^2 (d) $96\sqrt{3}$ cm^2

[SSC, 2015]

46. ABC is a cyclic triangle and the bisectors of $\angle$ BAC, $\angle$ ABC and $\angle$ BCA meet the circle at P, Q and R respectively. Then the angle $\angle$ RQP is:

(a) $90^\circ + \frac{C}{2}$ (b) $90^\circ - \frac{A}{2}$

(c) $90^\circ + \frac{B}{2}$ (d) $90^\circ - \frac{B}{2}$

[SSC, 2015]

47. ABC is a triangle and the sides AB, BC and CA are produced to E, F and G respectively. If $\angle$ CBE = $\angle$ ACF = 130°, then the value of $\angle$ GAB is:

(a) 80° (b) 130°

(c) 90° (d) 100°

[SSC, 2015]

48. In a rhombus ABCD, $\angle$ A = 60° and AB = 12cm. Then the diagonal BD is:

(a) 6 cm (b) 12 cm

(c) 10 cm (d) $2\sqrt{3}$ cm

[SSC, 2015]

49. If $x + \frac{1}{x} = 1$, then the value of $\frac{x^2+3x+1}{x^2+7x+1}$ is:

(a) $\frac{3}{7}$ (b) 2

(c) 1 (d) $\frac{1}{2}$

[SSC, 2015]

50. The ratio of each interior angle to each exterior angle of a regular polygon is 3:1. The number of sides of the polygon is:

(a) 7 (b) 8

(c) 9 (d) 6

[SSC, 2015]

51. O is the incentre of ΔPQR and ∠QPR = 50°, then the measure of ∠QOR is:-

(a) 130° (b) 100°

(c) 125° (d) 115°

[SSC, 2015]

52. O is the circumcentre of ΔABC. If ∠BAC = 85° ∠BCA = 75°, the ∠OAC is equal to:-

(a) 70° (b) 50°

(c) 40° (d) 60°

[SSC, 2015]

53. ABCD is a cyclic quadrilateral. Diagnals AC and BD meets at P. If ∠APB = 110° and ∠CBD = 30°, then ∠ADB measures-

(a) 55° (b) 30°

(c) 80° (d) 70°

[SSC, 2015]

54. The internal bisectors of the ∠B and ∠C of the ∠ABC, interest at O. If ∠A = 100°, then the measure of ∠BOC is:-

(a) 130° (b) 140°

(c) 110° (d) 120°

[SSC, 2015]

55. Given that: ΔABC ~ ΔPQR, If $\frac{\text{area}(\Delta PQR)}{\text{area}(\Delta ABC)} = \frac{256}{441}$ and PR = 12cm. then AC is equal to?

(a) 16 cm (b) 15.75 cm

(c) 12 $\sqrt{2}$ cm (d) 15.5 cm

[SSC, 2015]

56. In ΔABC, D and E are two mid points of sides AB and AC respectively. If ∠BAC = 40° and ∠ABC = 65° then ∠CED is :-

(a) 75° (b) 125°

(c) 130° (d) 105°

[SSC, 2015]

57. Let C1 and C2 be the inscribed and circumscribed circles of a triangle with sides 3cm, 4cm, and 5cm then area of C1 is area of C2

(a) $\frac{9}{16}$ (b) $\frac{9}{25}$

(c) $\frac{4}{25}$ (d) $\frac{16}{25}$

[SSC, 2015]

58. G is the centrioid of ΔABC. The medians AD and BE intersect at right angles. If the lengths of AD and BE are 9 cm and 12 cm respectively; then the length of AB (in cm) is?

(a) 10 (b) 10.5

(c) 9.5 (d) 11

[SSC, 2015]

59. If the three angles of a triangle are: $(x+15^\circ), \left(\frac{6x}{5}+6^\circ\right)$ and $\left(\frac{2x}{3}+30^\circ\right)$ then the triangle is:

(a) Scalene (b) isocsceles

(c) right angled (d) equilateral

[SSC, 2015]

60. If the number of vertices, edges and faces of a rectangular parallelepiped are denoted by v, e and f respectively, the value of (v – e + f) is
(a) 4 (b) 2
(c) 1 (d) 0
[SSC, 2015]

61. If the measure of three angles of a triangle are in the ratio 2 : 3 : 5, then the triangle is:
(a) equilateral (b) isocsceles
(b) Obtuse angled (d) right angled
[SSC, 2015]

62. If the sum and difference of two angles are $\frac{22}{9}$ radian and 36° respectively, then the value of smaller angle in degree taking the value of Π as $\frac{22}{7}$ is:
(a) 60° (b) 48°
(c) 52° (d) 56°
[SSC, 2015]

63. Internal bisectors of ∠Q and ∠R of ΔPQR intersect at O. If ∠ROQ = 96° the the value of ∠RPQ is:
(a) 12° (b) 6°
(c) 36° (d) 24°
[SSC, 2015]

64. If D, E and F are the mid points of BC, CA and AB respectively of the ΔABC then the ratio of area of the parallelogram DEFB and area of the trapezium CAFD is:
(a) 1 : 3 (b) 1 : 2
(c) 3 : 4 (d) 2 : 3
[SSC, 2015]

65. a and b are two sides adjacent to the right angle of a right-angle triangle and p is the perpendicular drawn to the hypotenuse from the opposite vertex. Then p^2 is equal to:
(a) a^2+b^2 (b) $\frac{1}{a^2}+\frac{1}{b^2}$
(c) $\frac{a^2b^2}{a^2+b^2}$ (d) a^2-b^2
[SSC, 2014]

66. Two chords of lengths a metres and b metres subtend angles 60°and 90° at the centre of the circle, respectively. Which of the following is true?
(a) $b=\sqrt{2}a$ (b) $a=\sqrt{2}b$
(c) $a = 2b$ (c) $b = 2a$
[SSC, 2014]

67. In a $\Delta ABC, \angle A+\frac{1}{2}\angle B+\angle C=140°$, then ∠B is:
(a) 50° (b) 80°
(c) 40° (d) 60°
[SSC, 2014]

68. The radius of a circle is 6 cm. The distance of a point lying outside the circle from the centre is 10 cm. The length of the tangent drawn from the outside point to the circle is:
(a) 5 cm (b) 6 cm
(c) 7 cm (d) 8 cm
[SSC, 2014]

69. If $ABCD$ is a cyclic quadrilateral in which $\angle A=4x°$, $\angle B=7x°$, $\angle C=5y°$, $\angle D=y°$, then $x:y$ is:
(a) 3:4 (b) 4:3
(c) 5:4 (d) 4:5
[SSC, 2014]

70. G is the centroid of the equilateral ΔABC. If AB = 10 cm, then length of AG is:
(a) $\frac{5\sqrt{3}}{3}$ cm (b) $\frac{10\sqrt{3}}{3}$ cm
(c) $5\sqrt{3}$ cm (d) $10\sqrt{3}$ cm
[SSC, 2014]

71. Two chords AB and CD of a circle with centre O, intersect each other at P. If $\angle AOD=100°$ and $\angle BOC=70°$, then the value of $\angle APC$ is
(a) 80° (b) 75°
(c) 85° (d) 95°
[SSC, 2014]

72. $ABCD$ is a cyclic quadrilateral and AD is a diameter. If $\angle DAC=55°$, then value of $\angle ABC$ is:
(a) 55° (b) 35°
(c) 145° (d) 125°
[SSC, 2014]

73. In ΔABC a straight line parallel to BC intersects AB and AC at D and E, respectively. If $AB = 2AD$, then $DE:BC$ is:
(a) 2:3 (b) 2:1
(c) 1:2 (d) 1:3
[SSC, 2014]

74. ABC is an isosceles triangle such that $AB = AC$ and AD is the median to the base BC with $\angle ABC=35°$. Then $\angle BAD$ is:
(a) 35° (b) 55°
(c) 70° (d) 110°
[SSC, 2014]

75. A man goes 24 m due west and then 10 m due north. Now, the distance of him from the starting point is:
(a) 17 m (b) 26 m
(c) 28 m (d) 34 m
[SSC, 2014]

76. ABC is a right angled triangle, B being the right angle. Mid-points of BC and AC are respectively B' and A'. Area of $\Delta A'B'C$ is

(a) $\frac{1}{2} \times$ area of ΔABC

(b) $\frac{2}{3} \times$ area of ΔABC

(c) $\frac{1}{4} \times$ area of ΔABC

(d) $\frac{1}{8} \times$ area of ΔABC

[SSC, 2014]

77. 'O' is the centre of the circle, AB is a chord of the circle, OM $\perp$ AB. If AB = 20 cm, OM = $2\sqrt{11}$ cm, then radius of the circle is

(a) 15 cm (b) 12 cm
(c) 10 cm (d) 11 cm

[SSC, 2014]

78. If the angles of a triangle ABC are in the ratio 2 : 3 : 1, then the angles $\angle A, \angle B, \angle C$ is

(a) $\angle A = 60°, \angle B = 90°, \angle C = 30°$

(b) $\angle A = 40°, \angle B = 120°, \angle C = 20°$

(c) $\angle A = 20°, \angle B = 60°, \angle C = 60°$

(d) $\angle A = 45°, \angle B = 90°, \angle C = 45°$

[SSC, 2014]

79. In ΔABC, $\angle ABC = 70°$, $\angle BCA = 40°$. O is the point of intersection of the perpendicular bisectors of the sides, then the angle $\angle BOC$ is

(a) 100° (b) 120°
(c) 130° (d) 140°

[SSC, 2014]

80. If the measures of the sides of triangle are $(x^2 - 1)$, $(x^2 + 1)$ and $2x$ cm, then the triangle would be

(a) equilateral (b) acute-angled
(c) isosceles (d) right-angled

[SSC, 2014]

81. If angle bisector of a triangle bisect the opposite side, then what type of triangle is?

(a) Right angled (b) Scalene
(c) Similar (d) Isosceles

[SSC, 2014]

82. If each angle of a triangle is less than the sum of the other two, then the triangle is

(a) obtuse angled (b) right angled
(c) acute angled (d) equilateral

[SSC, 2014]

83. A, B, C are three points on the circumference of a circle and if $\overline{AB} = \overline{AC} = 5\sqrt{2}$ cm and $\angle BAC = 90°$, find the radius.

(a) 10 cm (b) 5 cm
(c) 20 cm (d) 15 cm

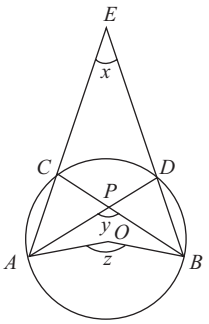
[SSC, 2014]

84. In ΔPQR, S and T are points on sides PR and PQ respectively such that $\angle PQR = \angle PST$. If PT = 5 cm, PS = 3 cm and TQ = 3 cm, then length of SR is

(a) 5 cm (b) 6 cm
(c) $\frac{31}{3}$ cm (d) $\frac{41}{3}$cm

[SSC, 2014]

85. The length of a tangent from an external point to a circle is $5\sqrt{3}$ unit. If radius of the circle is 5 units, then the distance of the point from the circle is



(a) 5 units (b) 15 units
(c) –5 units (d) –15 units

[SSC, 2014]

86. In ΔABC, $\angle C$ is an obtuse angle. The bisectors of the exterior angles at A and B meet BC and AC produced at D and E respectively. If AB = AD = BE, then $\angle ACB =$

(a) 105° (b) 108°
(c) 110° (d) 135°

[SSC, 2014]

87. ABC is an equilateral triangle and O is its circumcentre, then the $\angle AOC$ is

(a) 100° (b) 110°
(c) 120° (d) 130°

[SSC, 2014]

88. ABCD is a cyclic quadrilateral. The side AB is extended to E in such a way that BE = BC. If $\angle ADC = 70°$, $\angle BAD = 95°$, then $\angle DCE =$ is

(a) 140° (b) $120°$
(c) 165° (d) 110°

[SSC, 2014]

89. If the sum of interior angles of a regular polygon is equal to two times the sum of exterior angles of that polygon, then the number of sides of that polygon is

(a) 5 (b) 6
(c) 7 (d) 8

[SSC, 2014]

90. Two circles are of radii 7 cm and 2 cm, their centres being 13 cm apart. Then the length of direct common tangent to the circles between the points of contact is

(a) 12 cm (b) 15 cm
(c) 10 cm (d) 5 cm

[SSC, 2014]

91. In two similar triangles ABC and MNP, if AB = 2·25 cm MP = 4·5 cm and PN = 7·5 cm and m ∠ABC = m ∠MPN, then the length of side $\overrightarrow{BC}$, in cm, is

(a) 3·5 (b) 4·5
(c) 3·75 (d) 4·75

[SSC, 2014]

92. Given an equilateral ΔABC, D, E, F are the mid-points of AB, BC and AC respectively. Then the quadrilateral BEFD is exactly a

(a) trapezium (b) rhombus
(c) square (d) rectangle

[SSC, 2014]

93. If ABCD is a cyclic parallelogram, then the ∠A is

(a) 90° (b) 100°
(c) 60° (d) 80°

[SSC, 2014]

94. AC is a chord of circle whose centre is at O. If B is any point on the arc AC and ∠OCA = 20°, then the magnitude of ∠ABC is

(a) 140° (b) 110°
(c) 100° (d) 40°

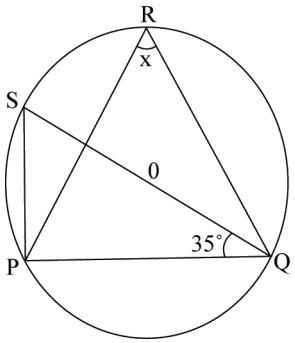
[SSC, 2014]

95. The co-ordinates of the vertices of a right-angled triangle are P(3, 4), Q(7, 4) and R(3, 8), the right-angle being at P. The co-ordinates of the orthocenter of Δ PQR are

(a) (3, 8) (b) 5,6)
(c) (3, 4) (d) (7, 4)

[SSC, 2014]

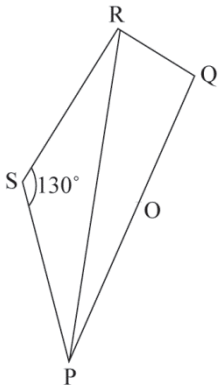
96. In the given figure, O is the centre. Then x is equal to



(a) 65° (b) 60°
(c) 45° (d) 55°

[SSC, 2014]

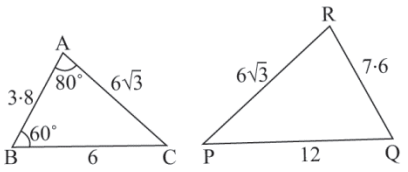
97. In the given figure, POQ is a diameter and PQRS is a cyclic quadrilateral. If ∠PSR = 130°, then the value of ∠RPQ is



(a) 35° (b) 30°
(c) 40° (d) 45°

[SSC, 2014]

98. In the given figures, the lengths of the sides of ΔABC and Δ PQR are given and they are given in same units. Also ∠A and ∠B are given. The value of ∠P is



(a) 40° (b) 42°
(c) 36° (d) 38°

[SSC, 2014]

99. In a $\Delta ABC, \angle A + \angle B = 180°$. Find the value of ∠ A.

(a) 30° (b) 34°
(c) 36° (d) 40°

[SSC, 2014]

100. In ΔABC, if AD $\perp$ BC, then $AB^2 + CD^2$ is equal to
(a) $2BD^2$ (b) $2AC^2$
(c) $BD^2 + AC^2$ (d) None of these

[SSC, 2014]

101. ABC is an equilateral triangle and CD is the internal bisector of $\angle C$. If DC is produced to E such that AC = CE, then $\angle CAE$ is equal to
(a) 30° (b) 15°
(c) 45° (d) 75°

[SSC, 2014]

102. $\angle ACB$ is an angle in the semicircle of diameter AB = 5 cm and AC : BC = 3 : 4. The area of the triangle ABC is
(a) 12 sq. cm (b) 6 sq. cm
(c) $6\sqrt{2}$ sq. cm (d) 4 sq. cm

[SSC, 2014]

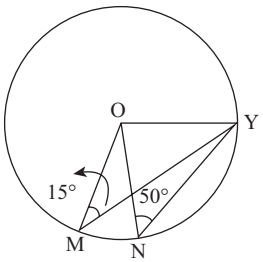
103. A, B and C are three points on a circle such that the angles subtended by the chords AB and AC at the centre O are 90° and 110° respectively. Further suppose that the center 'O' lies in the interior of $\angle$ BAC. Then $\angle BAC$ is
(a) 160° (b) 20°
(c) 40° (d) 80°

[SSC, 2014]

104. If the lengths of the sides AB, BC and CA of a triangle ABC are 10 cm, 8 cm and 6 cm respectively and if M is the mid-point of BC and MN ||AB to cut AC at N, then the area of the trapezium ABMN is equal to
(a) 12 sq. cm (b) 16 sq. cm
(c) 18 sq. cm (d) 20 sq. cm

[SSC, 2014]

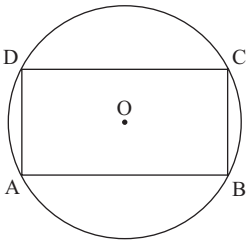
105. In the given figure, $\angle ONY = 50°$ and $\angle OMY = 15°$. Then the value of the $\angle MON$ is



(a) 20° (b) 70°
(c) 30° (d) 40°

[SSC, 2014]

106. In a cyclic quadrilateral
$\angle A + \angle C = \angle B + \angle D = ?$



(a) 90° (b) 180°
(c) 270° (d) 360°

[SSC, 2014]

107. $\angle A$, $\angle B$, $\angle C$ are three angles of a triangle. If $\angle A - \angle B = 15°$, $\angle B - \angle C = 30°$, then $\angle A$, $\angle B$ and $\angle C$ are:
(a) 80°, 60°, 40° (b) 70°, 50°, 60°
(c) 80°, 65°, 35° (d) 80°, 55°, 45°

[SSC, 2013]

108. If ABC is an equilateral triangle and D is a point on BC such that $AD \perp BC$, then:
(a) $AB:BD = 1:1$ (b) $AB:BD = 1:2$
(c) $AB:BD = 2:1$ (d) $AB:BD = 3:2$

[SSC, 2013]

109. ΔABC is an isosceles triangle and $\overline{AB} = \overline{AC} = 2a$ units, $\overline{BC} = a$ units. Draw $\overline{AD} \perp \overline{BC}$ and find the length of $\overline{AD}$.
(a) $\sqrt{15}a$ units (b) $\sqrt{\frac{15}{2}}a$ units
(c) $\sqrt{17}a$ units (d) $\sqrt{\frac{17}{2}}a$ units

[SSC, 2013]

110. All sides of a quadrilateral $ABCD$ touch a circle. If AB = 6 cm, BC = 7.5 cm, CD = 3 cm, then DA is:
(a) 3.5 cm (b) 4.5 cm
(c) 2.5 cm (d) 1.5 cm

[SSC, 2013]

111. In a right-angled triangle, the product of two sides is equal to half of the square of the third side, i.e., hypotenuse. One of the acute angles must be:
(a) 60° (b) 30°
(c) 45° (d) 15°

[SSC, 2013]

112. If two concentric circles are of radii 5 cm and 3 cm, then the length of the chord of the larger circle which touches the smaller circle is:

(a) 6 cm (b) 7 cm
(c) 10 cm (d) 8 cm

[SSC, 2013]

113. Inside a square $ABCD$, ΔBEC is an equilateral triangle. If CE and BD intersect at O, then $\angle BOC$ is equal to:
(a) 60° (b) 75°
(c) 90° (d) 120°

[SSC, 2013]

114. A point D is taken from the side BC of a right-angled triangle ABC, where AB is hypotenuse. Then:
(a) $AB^2 + CD^2 = BC^2 + AD^2$
(b) $CD^2 + BD^2 = 2AD^2$
(c) $AB^2 + AC^2 = 2AD^2$
(d) $AB^2 = AD^2 + BD^2$

[SSC, 2013]

115. Let C be a point on a straight line AB. Circles are drawn with diameters AC and AB. Let P be any point on the circumference of the circle with diameter AB. If AP meets the other circle at Q, then:
(a) $QC \parallel PB$
(b) QC is never parallel to PB
(c) $QC = \frac{1}{2}PB$
(d) $QC \parallel PB$ and $QC = \frac{1}{2}PB$

[SSC, 2013]

116. An isosceles triangle ABC is right-angled at B. D is a point inside the triangle ABC. P and Q are the feet of the perpendiculars drawn from D on the sides AB and AC, respectively of ΔABC. If $AP = a$ cm, $AQ = b$ cm and $\angle BAD = 15°$, then find sin 75°.
(a) $\frac{2b}{\sqrt{3}a}$ (b) $\frac{a}{2b}$
(c) $\frac{\sqrt{3}a}{2b}$ (d) $\frac{2a}{\sqrt{3}b}$

[SSC, 2013]

117. Each interior angle of a regular octagon in radians is:
(a) $\frac{\pi}{4}$ (b) $\frac{3\pi}{4}$
(c) $\frac{2\pi}{3}$ (d) $\frac{1}{3}\neq$

[SSC, 2013]

118. D and E are two points on the sides AC and BC, respectively of ΔABC such that DE = 18 cm, CE = 5 cm and $\angle DEC = 90°$. If tan ($\angle ABC$) = 3.6, then find AC:CD.
(a) BC:2CE (b) 2CE:BC
(c) 2CD:CE (d) CE:2BC

[SSC, 2013]

119. D is a point on the side BC of a triangle ABC such that $AD \perp BC$. E is a point on AD for which AE:ED = 5:1. If $\angle BAD = 30°$ and tan ($\angle ACB$) = 6 tan ($\angle DBE$), then find $\angle ACB$.
(a) 30° (b) 45°
(c) 60° (d) 15°

[SSC, 2013]

120. If the internal bisectors of $\angle ABC$ and $\angle ACB$ of ΔABC meet at O and also $\angle BAC = 80°$, then $\angle BOC$ is equal to:
(a) 50° (b) 160°
(c) 40° (d) 130°

[SSC Assistant Grade III, 2013]

121. O is the incenter of ΔABC. If $\angle BOC = 116°$, then $\angle BAC$ is:
(a) 42° (b) 62°
(c) 58° (d) 52°

[SSC Assistant Grade III, 2013]

122. Inside a triangle ABC, a straight line parallel to BC intersects AB and AC at the points P and Q, respectively. If $AB = 3PB$, then PQ:BC is:
(a) 1:3 (b) 3:4
(c) 1:2 (d) 2:3

[SSC Assistant Grade III, 2013]

123. O is the circumcentre of ΔABC. If $\angle BAC = 85°$, $\angle BCA = 75°$, then $\angle OAC$ is equal to:
(a) 70° (b) 60°
(c) 80° (d) 100°

[SSC Assistant Grade III, 2012]

124. The distance between the centres of the two circles with radii 4 cm and 9 cm is 13 cm. The length of the direct common tangent (between two points of contact) is:
(a) 13 cm (b) $\sqrt{153}$ cm
(c) 12 cm (d) 18 cm

[SSC Assistant Grade III, 2012]

125. The external bisector of $\angle ABC$ of ΔABC intersects the straight line through A and parallel to BC at the point D. If $\angle ABC = 50°$, then measure of $\angle ADB$ is:
(a) 65° (b) 55°
(c) 40° (d) 20°

[SSC Assistant Grade III, 2012]

126. AB is a diameter of a circle with centre at O. DC is a chord of it such that $DC \parallel AB$. If $\angle BAC = 20°$, then $\angle ADC$ is equal to:

(a) 120° (b) 110°
(c) 115° (d) 100°

[SSC Assistant Grade III, 2012]

127. The tangents drawn at P and Q on the circumference of a circle intersect at A. If $\angle PAQ = 68°$, then the measure of the $\angle APQ$ is:
(a) 56° (b) 68°
(c) 28° (d) 34°

[SSC Assistant Grade III, 2012]

128. If the incentre of an equilateral triangle lies inside the triangle and its radius is 3 cm, then the side of the equilateral triangle is:
(a) $9\sqrt{3}$ cm (b) $6\sqrt{3}$ cm
(c) $3\sqrt{3}$ cm (d) 6 cm

[SSC, 2012]

129. Suppose ΔABC be a right-angled triangle where $\angle A = 90°$ and $AD \perp BC$. If Area $(\Delta ABC) = 40$ cm^2, Area $(\Delta ACD) = 10$ cm^2 and $AC = 9$ cm, then the length of BC is:
(a) 12 cm (b) 18 cm
(c) 4 cm (d) 6 cm

[SSC, 2012]

130. Two circles touch each other externally at P. AB is a direct common tangent to the two circles, A and B are points of contact and $\angle PAB = 35°$. Then $\angle ABP$ is:
(a) 35° (b) 55°
(c) 65° (d) 75°

[SSC, 2012]

131. The length of the common chord of two intersecting circles is 24 cm. If the diameters of the circles are 30 cm and 26 cm, then the distance between the centres in cm is:
(a) 13 (b) 14
(c) 15 (d) 16

[SSC, 2012]

132. In ΔABC, D and E are points on AB and AC respectively such the $DE \parallel BC$ and DE divides the ΔABC into two parts of equal areas. Then ratio $AD:BD$ is:
(a) 1:1 (b) $1:\sqrt{2}-1$
(c) $1:\sqrt{2}$ (d) $1:\sqrt{2}+1$

[SSC, 2012]

133. X and Y are centres of circles of radii 9 cm and 2 cm respectively, $XY = 17$ cm. Z is the centre of a circle of radius r cm which touches the above circles externally. Given that $\angle XZY = 90°$, the value of r is:
(a) 13 cm (b) 6 cm
(c) 9 cm (d) 8 cm

[SSC, 2012]

134. I is the incentre of a triangle ABC. If $\angle ABC = 65°$ and $\angle ACB = 55°$, then the value of $\angle BIC$ is:
(a) 130° (b) 120°
(c) 140° (d) 110°

[SSC, 2012]

135. If the radii of two circles be 6 cm and 3 cm and the length of the transverse common tangent be 8 cm, then the distance between the two centres is:
(a) $\sqrt{145}$ cm (b) $\sqrt{140}$ cm
(c) $\sqrt{150}$ cm (d) $\sqrt{135}$ cm

[SSC, 2012]

136. The ratio between the numbers of sides of two regular polygons is 1:2 and the ratio between their interior angles is 2:3. The number of sides of these polygons is respectively:
(a) 6 (b) 8
(c) 4 (d) 5

[SSC, 2012]

137. The angles of a triangle are in Arithmetic Progression. The ratio of the least angle in degrees to the number of radians in the greatest angle is $60:\pi$ .The angles in degrees are:
(a) 30°, 60°, 90° (b) 35°, 55°, 90°
(c) 40°, 50°, 90° (d) 40°, 55°, 85°

[SSC, 2012]

138. The ratio between the adjacent angles of a parallelogram are 2:3. The smallest angle of a quadrilateral is equal to the half of the smallest angle of a parallelogram. The highest angle of a quadrilateral is 4 times greater than its smallest angle. What is the sum of the highest angle of a quadrilateral and the smallest angles of a parallelogram?
(a) 252° (b) 226°
(c) 144° (d) 180°

[Union Bank of India PO, 2011]

139. One of the angles of a triangle is two-thirds angle of sum of adjacent angles of parallelogram. Remaining angles of the triangle are in ratio 5:7 respectively. What is the value of second largest angle of the triangle?
(a) 25°
(b) 40°
(c) 35°
(d) Cannot be determined

[Corporation Bank PO, 2011]

140. Smallest angle of a triangle is equal to two-thirds the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is 3:4:5:6. Largest angle of the triangle is twice its smallest angle. What is

the sum of second largest angle of the triangle and largest angle of the quadrilateral?

(a) 160º (b) 180º
(c) 190º (d) 170º

[Bank of Baroda PO Examination, 2011]

141. A, B, C are three points on a circle. The tangent at A meets BC produced at T, $\angle BTA = 40°$, $\angle CAT = 44°$. The angle subtended by BC at the centre of the circle is:

(a) 84° (b) 92°
(c) 96° (d) 104°

[SSC, 2011]

142. If the length of a chord of a circle at a distance of 12 cm from the centre is 10 cm, then the diameter of the circle is:

(a) 13 cm (b) 15 cm
(c) 26 cm (d) 30 cm

[SSC, 2011]

143. In ΔABC, P and Q are the middle points of the sides AB and AC respectively. R is a point on the segment PQ such that $PR:RQ = 1:2$. If $PR = 2$ cm, then $BC =$

(a) 4 cm (b) 2 cm
(c) 12 cm (d) 6 cm

[SSC, 2011]

144. If O is the circumcentre of ΔABC and $\angle OBC = 35°$, then the $\angle BAC$ is equal to:

(a) 55° (b) 110°
(c) 70° (d) 35°

[SSC, 2011]

145. If I is the incentre of ΔABC and $\angle BIC = 135°$, then ΔABC is:

(a) acute angled (b) equilateral
(c) right angled (d) obtuse angled

[SSC, 2011]

146. The ratio of the adjacent angles of a parallelogram is 7:8. Also, the ratio of the angles of quadrilateral is 5:6:7:12. What is the sum of the smaller angle of the parallelogram and the second largest angle of the quadrilateral?

(a) 168º (b) 228º
(c) 156º (d) 224º
(e) None of these

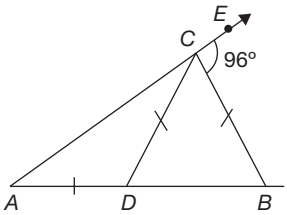
[IOB PO, 2011]

147. One of the angles of a triangle is two-thirds of the sum of the adjacent angles of parallelogram. Remaining angles of the triangle are in the ratio 5:7. What is the value of the second largest angle of the triangle?

(a) 25° (b) 40°
(c) 35° (d) Cannot be determined
(e) None of these

[Corporation Bank PO, 2011]

148. In the figure (not drawn to scale) give below, if AD = DC = BC and $\angle BCE = 96°$, then $\angle DBC$ is:



(a) 32° (b) 84°
(c) 64° (d) 96°

[SSC CGL Tier-I (CBE), 2011]

149. The angles of a quadrilateral are in the ratio of 2:4:7:5. The smallest angle of the quadrilateral is equal to the smallest angle of a triangle. One of the angles of the triangle is twice the smallest angle of the triangle. What is the second largest angle of the triangle?

(a) 80° (b) 60°
(c) 120° (d) Cannot be determined

[CBI (PO), 2010]

150. The ratio between the angles of a quadrilateral is 3:4:6:5. Two-thirds the largest angle of the quadrilateral is equal to the smaller angle of a parallelogram? What is the value of adjacent angle of the parallelogram?

(a) 120° (b) 110°
(c) 100° (d) 130°

[OBC PO, 2010]

151. The largest and the second largest angles of a triangle are in the ratio of 3:2, respectively. The smallest angle is 20% of the sum of the largest and the second largest angles. What is the sum of the smallest and the second largest angles?

(a) 80º (b) 60º
(c) 100º (d) 90º

[Bank of Baroda PO, 2010]

152. What is the length of the radius of the circum-circle of the equilateral triangle, the length of whose side is $6\sqrt{3}$ cm?

(a) $6\sqrt{3}$ cm (b) 6 cm
(c) 5.4 cm (d) $3\sqrt{6}$ cm

[SSC, 2010]

153. The two sides of a right triangle containing the right angle measure 3cm and 4cm. The radius of the incircle of the triangle is:

(a) 3.5 cm (b) 1.75 cm
(c) 1 cm (d) 0.875 cm

154. In a circular lawn, there is a 16 m long path in the form of a chord. If the path is 6 m away from the centre of the lawn, then find the radius of the circular lawn.

(a) 16 m (b) 6 m
(c) 10 m (d) 8 m

155. In a triangle ABC, the length of the sides AB, AC and BC are 3, 5 and 6 cm respectively. If a point D on BC is drawn such that the line AD bisects the angle A internally, then what is the length of BD?

(a) 2 cm (b) 2.25 cm
(c) 2.5 cm (d) 3 cm

156. In a triangle ABC, $\angle A = x°$, $\angle B = y°$ and $\angle C = (y + 20)°$. If $4x - y = 10$, then the triangle is:

(a) Right-angle (b) Obtuse-angled
(c) Equilateral (d) None of these

157. If the sides of a right triangle are x, $x + 1$ and $x - 1$, then the hypotenuse:

(a) 5 (b) 4
(c) 1 (d) 0

158. If P and Q are the mid points of the sides CA and GB respectively of a triangle ABC, right-angled at C. Then the value of $4\ (AQ^2 + BP^2)$ is equal to:

(a) $4\ BC^2$ (b) $5\ AB^2$
(c) $2\ AC^2$ (d) $2\ BC^2$

159. In a quadrilateral $ABCD$, $\angle B = 90°$ and $AD^2 = AB^2 + BC^2 + CD^2$, then $\angle ACD$ is equal to:

(a) 90° (b) 60°
(c) 30° (d) None of these

160. $ABCD$ is a square, F is mid point of AB and E is a point on BC such that BE is one-third of BC. If area of $\Delta FBE = 108\ m^2$, then the length of AC is:

(a) 63 m (b) $36\sqrt{2}$ m
(c) $63\sqrt{2}$ m (d) $72\sqrt{2}$ m

161. We have an angle of $2\frac{1^\circ}{2}$. How big it will look through a glass that magnifies things three times?

(a) $2\frac{1^\circ}{2} \times 4$ (b) $2\frac{1^\circ}{2} \times 3$

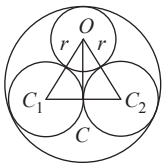
(c) $2\frac{1^\circ}{2} \times 2$ (d) None of these

162. Two circles with radii 'a' and 'b' respectively touch each other externally. Let 'c' be the radius of a circle that touches these two circle as well as a common tagent to the two circles. Then:

(a) $\frac{1}{\sqrt{a}} - \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{c}}$ (b) $\frac{1}{\sqrt{a}} - \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{c}}$

(c) $\frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} = \frac{1}{\sqrt{c}}$ (d) None of these

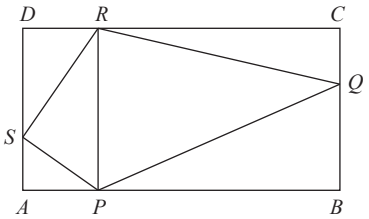
163. Two circles of unit radius touch each other and each of them touches internally a circle of radius two, as shown in the following figure. The radius of the circle which thouches all the three circles:



(a) 5 (b) $\frac{3}{2}$

(c) $\frac{2}{3}$ (d) None of these

164. $ABCD$ is a parallelogram P, Q, R and S are points on sides AB, BC, CD and DA respectively such that $AP = DR$. If the area of the parallelogram $ABCD$ is 16 cm^2, then the area of the quadrilateral $PQRS$ is:

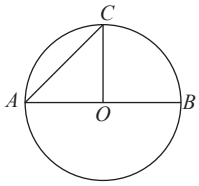


(a) 6 cm^2 (b) 6.4 cm^2
(c) 4 cm^2 (d) 8 cm^2

165. Let ABC be an acute-angled triangle and CD be the altitude through C. If $AB = 8$ and $CD = 6$, then the distance between the mid-points of AD and BC is:

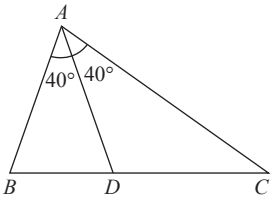
(a) 36 (b) 25
(c) 27 (d) 5

166. In the accompanying figure, AB is one of the diameters of the circle and OC is perpendicular to it through the centre O. If AC is $7\sqrt{2}$ cm, then what is the area of the circle in cm^2?



(a) 24.5 (b) 49
(c) 98 (d) 154

167. In the following figure, If $BC = 8$ cm, $AB = 6$ cm, AC 9 cm, then DC is equal to:



(a) 7 cm (b) 4.8 cm
(c) 7.2 cm (d) 4.5cm

168. With the vertices of a ΔABC as centers, three circles are described each touching the other two externally. If the sides of the triangle are 4, 6 and 8 cm respectively, then the sum of the radii of the three circles equals:

(a) 10 (b) 14
(c) 12 (d) 9

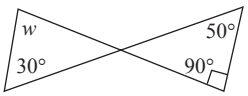
169. The sum of the interior angles of a polygon is 1620°. The number of sides of the polygon are

(a) 9 (b) 11
(c) 15 (d) 12

170. How many sides a regular polygon has with its interior angle eight times its exterior angle?

(a) 16 (b) 24
(c) 18 (d) 20

171. In the figure given below, what is the value of w?



(a) 100 (b) 110
(c) 120 (d) 130

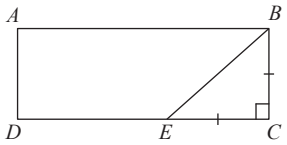
Notes

The diagram is not drawn to scale.

172. The radius of the circumcircle of an equilateral triangle of side 12 cm is:

(a) $\left(\frac{4}{3}\right)\sqrt{3}$ (b) $4\sqrt{2}$
(c) $4\sqrt{3}$ (d) 4
(e) None of above

173. In the diagram below, $ABCD$ is a rectangle. The area of isosceles right triangle BCE is 14, and $DE = 3EC$. What is the area of $ABCD$?



(a) 112 (b) 56
(c) 84 (d) $3\sqrt{28}$

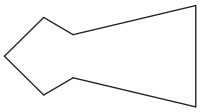
174. Find the distance of aperpendicular from the centre of a circle to the chord if the diameter of the circle is 30 cm and its chord is 24cm.

(a) 6 cm (b) 7 cm
(c) 9 cm (d) 10 cm

175. The medians AD, BE and CF of a triangle ABC intersect in G. Which of the following is true for any ΔABC?

(a) $GB + GC = 2GA$
(b) $GB + GC < GA$
(c) $GB + GC > GA$
(d) $GB + GC = GA$
(e) None of these

176. Find the sum of the degree measure of the internal angles in the polygon shown below.



(a) 600° (b) 720°
(c) 900° (d) 1080°
(e) 750°

177. A semi-circle is drawn with AB as its diameter. From C, a point on AB, a line perpendicular to AB is draw meeting the circumference of the semi-circle at D. Given that AC = 2 cm and CD = 6 cm the area of the semicircle (in cm^2) will be:

(a) 32π cm^2 (b) 50π cm^2
(c) 40.5π cm^2 (d) 81π cm^2
(e) undeterminable

178. An equilateral triangle BPC is drawn inside a square ABCD. What is the value of the angle APD in degrees:

(a) 75 (b) 90
(c) 12 (d) 135
(e) 150

179. The heights of two similar right-angled triangles Δ LMN and Δ OPQ are 48 cm and 36 cm. If OP = 12 cm, then LM is

(a) 12 cm (b) $\frac{10\sqrt{6}}{3}$
(c) 16 cm (d) 20 cm

ANSWER KEYS

EXERCISE–I

1. (c)	**2.** (b)	**3.** (a)	**4.** (d)	**5.** (a)	**6.** (d)	**7.** (a)	**8.** (b)	**9.** (a)	**10.** (c)	**11.** (b)	**12.** (d)
13. (c)	**14.** (b)	**15.** (b)	**16.** (d)	**17.** (a)	**18.** (c)	**19.** (a)	**20.** (a)	**21.** (b)	**22.** (c)	**23.** (c)	**24.** (a)
25. (a)	**26.** (d)	**27.** (b)	**28.** (c)	**29.** (a)	**30.** (c)	**31.** (a)	**32.** (b)	**33.** (c)	**34.** (a)	**35.** (d)	**36.** (b)
37. (d)	**38.** (b)	**39.** (c)	**40.** (d)	**41.** (a)	**42.** (d)	**43.** (b)	**44.** (b)	**45.** (b)	**46.** (c)	**47.** (b)	**48.** (c)
49. (b)	**50.** (a)	**51.** (d)	**52.** (c)	**53.** (a)	**54.** (c)	**55.** (d)	**56.** (d)	**57.** (a)	**58.** (d)	**59.** (a)	**60.** (c)
61. (c)	**62.** (d)	**63.** (a)	**64.** (c)	**65.** (a)	**66.** (a)	**67.** (a)	**68.** (c)	**69.** (b)	**70.** (c)	**71.** (d)	**72.** (d)
73. (a)	**74.** (b)	**75.** (c)	**76.** (b)	**77.** (a)	**78.** (d)	**79.** (c)	**80.** (c)	**81.** (a)	**82.** (d)	**83.** (c)	**84.** (c)
85. (b)	**86.** (d)	**87.** (a)	**88.** (c)	**89.** (c)	**90.** (c)	**91.** (a)	**92.** (b)	**93.** (d)	**94.** (a)	**95.** (d)	**96.** (c)
97. (b)	**98.** (d)	**99.** (a)	**100.** (c)	**101.** (a)	**102.** (d)	**103.** (b)	**104.** (c)				

EXERCISE–2

1. (c)	**2.** (c)	**3.** (b)	**4.** (b)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (d)	**11.** (b)	**12.** (d)
13. (b)	**14.** (c)	**15.** (b)	**16.** (c)	**17.** (c)	**18.** (d)	**19.** (b)	**20.** (c)	**21.** (b)	**22.** (c)	**23.** (b)	**24.** (b)
25. (a)	**26.** (c)	**27.** (b)	**28.** (a)	**29.** (d)	**30.** (c)	**31.** (c)	**32.** (c)	**33.** (b)	**34.** (b)	**35.** (a)	**36.** (b)
37. (d)	**38.** (b)	**39.** (a)	**40.** (b)	**41.** (d)	**42.** (c)	**43.** (c)	**44.** (b)	**45.** (a)	**46.** (d)	**47.** (d)	**48.** (b)
49. (d)	**50.** (b)	**51.** (d)	**52.** (a)	**53.** (c)	**54.** (b)	**55.** (b)	**56.** (d)	**57.** (c)	**58.** (a)	**59.** (d)	**60.** (b)
61. (d)	**62.** (c)	**63.** (a)	**64.** (d)	**65.** (c)	**66.** (a)	**67.** (b)	**68.** (d)	**69.** (b)	**70.** (b)	**71.** (d)	**72.** (c)
73. (c)	**74.** (b)	**75.** (b)	**76.** (c)	**77.** (b)	**78.** (a)	**79.** (d)	**80.** (d)	**81.** (d)	**82.** (c)	**83.** (b)	**84.** (c)
85. (a)	**86.** (b)	**87.** (c)	**88.** (a)	**89.** (b)	**90.** (a)	**91.** (c)	**92.** (b)	**93.** (a)	**94.** (b)	**95.** (c)	**96.** (d)
97. (c)	**98.** (a)	**99.** (b)	**100.** (c)	**101.** (d)	**102.** (b)	**103.** (d)	**104.** (c)	**105.** (b)	**106.** (b)	**107.** (c)	**108.** (c)
109. (b)	**110.** (d)	**111.** (c)	**112.** (d)	**113.** (b)	**114.** (a)	**115.** (d)	**116.** (c)	**117.** (b)	**118.** (c)	**119.** (c)	**120.** (d)
121. (d)	**122.** (d)	**123.** (a)	**124.** (c)	**125.** (a)	**126.** (b)	**127.** (a)	**128.** (b)	**129.** (b)	**130.** (b)	**131.** (b)	**132.** (b)
133. (b)	**134.** (b)	**135.** (a)	**136.** (c)	**137.** (a)	**138.** (d)	**139.** (c)	**140.** (b)	**141.** (d)	**142.** (c)	**143.** (c)	**144.** (a)
145. (c)	**146.** (a)	**147.** (c)	**148.** (c)	**149.** (b)	**150.** (c)	**151.** (d)	**152.** (b)	**153.** (c)	**154.** (c)	**155.** (b)	**156.** (a)
157. (a)	**158.** (b)	**159.** (a)	**160.** (b)	**161.** (d)	**162.** (c)	**163.** (c)	**164.** (d)	**165.** (d)	**166.** (d)	**167.** (b)	**168.** (d)
169. (b)	**170.** (c)	**171.** (b)	**172.** (c)	**173.** (a)	**174.** (c)	**175.** (c)	**176.** (c)	**177.** (b)	**178.** (e)	**179.** (c)	

EXPLANATORY ANSWERS

EXERCISE-I

1. (c) Let the measured of the required angle be x degree. Then, its supplement = $180 - x$

$$\text{Now, angle} = \frac{1}{3}\text{ (its supplement)}$$

$$x = \frac{1}{3}(180 - x)$$

$3x + x = 180° \Rightarrow x = 45°$.

2. (b) complement of $30°20' = 90° - (30°20') = 90° - (30° + 20')$

$= (89° - 30°) + (1° - 20')$

$= 59° + 60' - 20'$ [$\because$ $1° = 60'$]

$= 59° + 40' = 59°40'$.

3. (a) Since OP bisects $\angle BOC$,

$\therefore \angle BOC = 2\angle POC$

Again, OQ bisects $\angle AOC$, $\therefore$,

$\angle AOC = 2\angle QOC$

Since ray OC stands on line AB, $\therefore$,

$\angle AOC + \angle BOC = 180° \Rightarrow 2\angle QOC + 2\angle POC = 180°$

$\Rightarrow 2\angle QOC + \angle POC = 180°$

$\Rightarrow \angle QOC + \angle POC = 90°$

$\Rightarrow \angle POQ = 90°$.

The above sum can also be restated as follows; The angle between the bisectors of a linear pair of angles is a right angle.

4. (d) Through O, draw a line l parallel to both AB and CD. Then

$\angle 1 = 45°$ (alt. $\angle$S)

and $\angle 2 = 30°$ (alt. $\angle$S)

$\therefore \angle BOC = \angle 1 + \angle 2 = 45° + 30° = 75°$

So, $X = 360° - \angle BOC = 360° - 75° = 285°$

Hence $X = 285°$.

5. (a) AB || CD and t transversal intersects them at E and F

$\angle BEF + \angle EFD = 180°$ (co-interior angles)

$$\Rightarrow \frac{1}{2}\angle BEF + \frac{1}{2}\angle EFD = 180°$$

$\Rightarrow \angle BEF + \Delta EFD = 90°$

In $\angle EFG$

$\angle EFG + \angle FEG + \angle EGF = 180°$

$\therefore \angle EGF + 90° = 180°$

$\therefore \angle EGF = 90°$.

The above result can be restated as:

If two parallel lines are cut by a transversal, then the bisectors of the interior angles on the same side of the transversal intersect each other at right angles.

6. (d)

$\angle DCK = \angle FDG = 55°$ (corr. $\angle s$)

$\therefore \angle ACE = \angle DCK = 55°$ (vert. opp. $\angle s$)

So, $\angle AEC = 180° - (40° + 55°) = 85°$

$\therefore \angle HAB = \angle AEC = 85°$ (corr. $\angle s$)

Hence, $x = 85°$.

7. (a) Let, the measure of the required angled be $x°$.

Then, measure of its complement = $(90 - x)°$ measure of its supplement = $(180 - x)°$

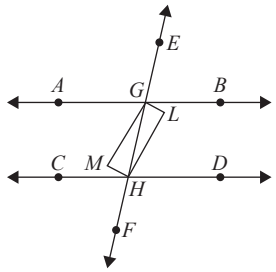
$6(90° - x) = 2(180° - x) - 12°$

$\Rightarrow 540° - 6x = 360° - 2x - 12°$

$\Rightarrow 4x = 192° \Rightarrow x = 48°$.

8. (b) $\angle AGH = \angle DHG$ (alt. int. angles)

$$\frac{1}{2}\angle AGH = \frac{1}{2}\angle DHG \Rightarrow \angle HGM = \angle GHL$$



Thus, lines GM and HL are intersected by a transversal GH at G and H respectively such that pair of alternate angles are equal, i.e., $\therefore \angle HGM = \angle GHL$

$\therefore GM \parallel HL$

Similarly, $GL \parallel HM$

So, $GMHL$ is a $\|_{gm}$.

Since $AB \parallel CD$ and EF is a transversal

$\therefore \angle BGH + \angle DHG = 180°$ [co-interior angles]

$$\therefore \frac{1}{2}\angle BGH + \frac{1}{2}\angle DHG = 90°$$

$\Rightarrow \angle LGH + \angle LHG = 90°$

But $\angle LGH + \angle LHG + \angle GLH = 180°$

$\therefore 90° + \angle GLH = 180° \Rightarrow \angle GLH = 90°$

Thus, in $\|_{gm}$ $GMHL$, we have $\angle GLH = 90°$

Hence, $GMHL$ is a rectangle.

9. (a) CD || AB (Given)

